

Qn.1. What is a recursive function, and how does it differ from an iterative solution?

A recursive function is a function that calls itself to solve a problem by breaking it into smaller sub- problems.

Recursive	Iterative
Function calls itself to solve a smaller sub-problem	Uses loops (for / while) to repeat tasks
Requires a base case to stop recursion	Loop continues until a condition is false
Uses more memory because of call stack	Uses less memory, only loop variables
Slower for large problems due to repeated calls	Usually faster and more efficient
Good for problems like trees, graphs, divide-and-conquer	Good for simple, repeated calculations

Qn.2. Why is it important to have a base case in a recursive function?

A base case is important in a recursive function because it stops the recursion. Without it, the function would call itself every time and eventually crash with a stack overflow.

Qn.3. What is a context manager in python?

A context manager in Python is a special construct that sets up a context for a block of code and cleans up automatically when the block ends. It is usually used with the with statement.

Syntax: with <context_manager> as <variable>:

Qn.4. What is the purpose of the with statement in Python?

The with statement in Python is used to simplify resource management. Its main purpose is to ensure that resources are properly acquired and released, even if an error occurs.

Qn.5. How does the with statement help in managing resources like files or network connections?

The with statement manages resources like files or network connections safely. It automatically calls `__enter__()` on start and `__exit__()` on end. This ensures proper cleanup, prevents resource leaks, and makes code cleaner.

Qn.6. What is the role of the enter and exit methods in a context manager, and how do they work together when using the with statement in Python?

`__enter__()` runs when the with block starts to set up the resource, and `__exit__()` runs when the block ends to clean it up.

When Python executes a with statement:

- Calls `__enter__()` → sets up the resource
- Runs the code inside the block
- Calls `__exit__()` → cleans up the resource automatically

Qn.7. What are the different file modes in Python (r, w, a, rb, wb)?

r- Read mode (opens a file for reading. File must exist.)

w- Write mode (opens a file for writing. Creates a new file or overwrites an existing file.)

a- Append mode (opens a file for writing at the end. Creates a new file if it doesn't exist.)

rb- Read binary (opens a file for reading in binary mode.)

wb- Write binary (opens a file for writing in binary mode. Creates or overwrites a file.)

Qn.8. What is the difference between `read()`, `readline()`, and `readlines()` methods?

`read()` → Reads the entire content of the file at once

`readline()` → Reads one line at a time from the file

`readlines()` → Reads all lines and returns them as a list of lines

Qn.9. What is a CSV file, and why is it widely used?

CSV (Comma-Separated Values) file is a plain text file where data is stored in rows and columns, with each value separated by a comma.

It is widely used because,

- Simple and lightweight, easy to read and write.
- Compatible with many applications like Excel, databases, and Python.
- Ideal for storing and exchanging tabular data between systems.

Qn.10. How does Python's csv module help in reading and writing CSV files?

Python's csv module provides tools to read from and write to CSV files easily.

- `csv.reader()` allows you to read CSV files row by row as lists.
- `csv.writer()` allows you to write data into CSV files row by row.
- It handles commas, quotes, and special characters automatically, making CSV processing safer and simpler.

Qn.11. What is JSON, and how is it different from Python dictionaries?

JSON (JavaScript Object Notation) is a lightweight, text-based format for storing and exchanging data. It represents data as key-value pairs, arrays, strings, numbers, booleans, or null.

Difference from Python dictionaries:

- JSON is a string format used for data exchange between programs or over the web.
- Python dictionaries are in-memory data structures used within Python programs.
- JSON can be converted to/from dictionaries using `json.loads()` and `json.dumps()` in Python.